

Assignment Title: AI Assignment

**Option H: An AI Tutor for Personalized
Student Learning Recommendations**

Roll No: 2K23/CSE/144

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Assignment 1: AI System Specification

Topic: AI Tutor for Personalized Student Learning Recommendations (Option H)

Task 1: PEAS Specification

Performance Measure:

- Learning gain: improvement in assessment scores between a pre-test and post-test on the same topic.
- Recommendation relevance: percentage of recommended exercises that target a skill gap the student actually has, as verified against their performance history.
- Engagement rate: proportion of recommended sessions that the student starts and completes without abandoning.
- Time-to-mastery: average number of practice attempts or sessions needed before a student reaches a mastery threshold (e.g. 90% accuracy) on a topic.
- Retention rate: accuracy on a topic when re-tested after a delay (e.g. two weeks), measuring whether learning gains persist.

Environment: A web or mobile e-learning platform used by individual students, typically at home or in a classroom, with variable internet connectivity and session lengths. Key external factors include the student's curriculum/syllabus, their prior knowledge level, available study time, device type, and occasional distractions or interruptions during a session.

Actuators:

- Content recommendation engine that selects the next lesson, video, or reading material.
- Adaptive quiz/exercise generator that adjusts question difficulty in real time.
- Feedback and hint display that shows explanations, worked solutions, or encouragement messages.
- Progress dashboard updates visible to the student (and optionally parents/teachers).
- Notification/reminder system that prompts the student to revisit weak topics or maintain a study streak.

Sensors:

- Quiz and assignment responses, including correctness and chosen options.
- Timing data: time spent per question, per lesson, and idle time between actions.
- Clickstream/navigation data showing which materials the student opens, skips, or revisits.
- Text or voice input from chat-based help requests, if the tutor supports conversational queries.
- Self-reported confidence ratings collected after exercises (e.g. a 1-5 scale).

Task 2: Environment Classification

Fully Observable vs. Partially Observable

Choice: Partially Observable. Justification: The system can only observe the student's behavior (answers, timing, clicks), not their actual internal knowledge state, confusion, or motivation, so it must infer hidden mental states from indirect evidence.

Single-agent vs. Multi-agent

Choice: Single-agent. Justification: From the perspective of the tutoring agent, the student is part of the environment rather than another cooperating or competing agent, and there is typically one tutor instance per student session making the decisions.

Deterministic vs. Stochastic

Choice: Stochastic. Justification: The same recommendation can lead to different outcomes for the same student on different days, since learning, attention, and performance are influenced by many unobservable and variable factors.

Episodic vs. Sequential

Choice: Sequential. Justification: Each recommendation affects the student's future knowledge state and the data available for future decisions, so current actions have long-term consequences rather than being independent of past and future episodes.

Static vs. Dynamic

Choice: Dynamic. Justification: The student's knowledge, fatigue, and motivation change continuously over time even while the agent is deciding what to recommend next, so the environment does not wait for the agent's action.

Discrete vs. Continuous

Choice: Discrete. Justification: Interactions occur as a finite sequence of distinct events, such as answering a question, completing a lesson, or receiving a recommendation, even though some underlying measurements like time are continuous.

Task 3: Critical Analysis

1. Greatest technical challenge

Of the six dimensions, partial observability poses the greatest technical challenge for an AI engineer building this system. The agent never has direct access to what a student actually understands; it must estimate this hidden knowledge state purely from proxies such as quiz scores, response times, and click patterns, which are noisy and can be misleading (a student might guess correctly, or answer slowly due to a distraction rather than confusion). Building an accurate model of student knowledge under this uncertainty, often using techniques like Bayesian Knowledge Tracing or deep knowledge tracing models, is considerably harder than handling the sequential or dynamic aspects, which can be addressed with more standard techniques such as reinforcement learning over time-indexed states.

2. Utility function and trade-off

A useful utility function for this agent balances learning gain against frustration risk: $U = w_1 * \text{learning_gain} - w_2 * \text{frustration_risk}$, where `learning_gain` is the estimated improvement in mastery from a recommended activity, and `frustration_risk` estimates the probability that the activity is too difficult and will cause the student to disengage. If the weight w_2 is doubled, the agent becomes far more conservative: it will increasingly favor easier or more familiar exercises, reduce the pace at which it introduces new or challenging material, and possibly recommend more review sessions even at the cost of slower overall progress. Over many sessions this could improve retention and engagement for struggling students, but for

confident or advanced students it may make the tutor feel too easy, leading to boredom and reduced engagement from the opposite cause.

Structured Representation

The JSON object below represents the Task 1 and Task 2 outputs in a structured, machine-readable format.

```
{
  "system_name": "AI Tutor for Personalized Student Learning",
  "peas": {
    "performance_measures": [
      "learning_gain",
      "recommendation_relevance",
      "engagement_rate",
      "time_to_mastery",
      "retention_rate"
    ],
    "environment": "Web or mobile e-learning platform used by individual students with variable curricula, prior knowledge, study time, devices, and connectivity",
    "actuators": [
      "content_recommendation_engine",
      "adaptive_quiz_generator",
      "feedback_hint_display",
      "progress_dashboard",
      "notification_system"
    ],
    "sensors": [
      "quiz_responses",
      "response_timing_data",
      "clickstream_navigation_data",
      "chat_text_voice_input",
      "self_reported_confidence_ratings"
    ]
  },
  "environment_classification": {
    "observability": {
      "choice": "Partially Observable",
      "justification": "The agent cannot directly see the student's true knowledge state, only indirect behavioral evidence"
    },
    "agents": {
      "choice": "Single-agent",
      "justification": "One tutoring agent makes recommendations per student session, treating the student as part of the environment"
    },
    "determinism": {
      "choice": "Stochastic",
      "justification": "Identical recommendations can produce different learning outcomes due to unobservable factors like attention and motivation"
    },
    "episodicity": {
      "choice": "Sequential",
      "justification": "Each recommendation influences the student's future knowledge state and subsequent decisions"
    },
    "dynamics": {
      "choice": "Dynamic",
      "justification": "The student's knowledge and engagement change over time independent of the agent's actions"
    },
    "continuity": {
      "choice": "Discrete",
      "justification": "Interactions occur as a finite sequence of distinct events such as answering questions and completing lessons"
    }
  },
  "utility_function": "U = w1 * learning_gain - w2 * frustration_risk"
}
```